

Project Initialization and Planning Phase

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal covers an AI system that reads a student's own description of a study problem, detects the emotion behind it, and adapts its response accordingly — rather than giving the same flat guidance regardless of whether the student is frustrated, confused, curious, confident, or bored.

Project Overview

Field	Details
Objective	Detect the emotional state behind a student's description of a study problem and respond with guidance appropriately adapted to that state and their subject area, rather than one-size-fits-all help.
Scope	Text-based emotion classification across 5 classes, a trust-adaptive decision layer combining two independently-trained models, AI-generated supportive responses via LLM, opt-in session logging, and an analytics dashboard — delivered as a single Streamlit web app.

Problem Statement

Field	Details
Description	Existing self-study support (generic chatbots, static material, asynchronous instructor access) responds to the content of a student's question but not the emotion behind it, giving the same flat tone to a frustrated student as to a bored or curious one.
Impact	Left unaddressed, this contributes to disengagement during independent study (boredom) and abandonment when stuck (frustration), and gives no signal for moments when a student's confidence or curiosity could be reinforced instead of ignored.

Proposed Solution

Field	Details
Approach	Train two independent text classifiers — a lightweight BiLSTM and a fine-tuned BERT — on the merged emotion dataset, combine their outputs through a rule-based decision engine calibrated against real validation data (not a fixed "always trust the bigger model" rule), and use the resulting emotion + trust level to condition an LLM prompt for the final response, with a deterministic template fallback when AI generation is off or unavailable.
Key Features	- Dual-model detection (BiLSTM + BERT) with mixed-emotion support - Data-driven trust/decision engine (4-branch logic, grounded in a 770-row validation analysis) - LLM-based, field- and emotion-aware response generation (Gemini/OpenRouter), with template fallback - Opt-in session history + CSV logging

Field	Details
	- Plotly analytics dashboard (emotion distribution, emotional-journey timeline, per-field breakdown) - Graceful degradation when BERT or the LLM API is unavailable

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware — Computing Resources	CPU/GPU used for training and local development	Local dev: NVIDIA RTX 5070 Ti (BERT fine-tuning, CUDA 12.8) + Ryzen 9 9900X (BiLSTM, CPU-only by design). Public deployment: Streamlit Community Cloud, CPU-only, free tier.
Hardware — Storage	Space for trained model weights	BERT (~440MB) hosted on Hugging Face Hub, downloaded at runtime; BiLSTM weights small enough to commit directly (model.keras, tokenizer.pickle).
Software — Frameworks	Core frameworks	Streamlit 1.37.1, TensorFlow (tensorflow-cpu 2.16.1), PyTorch + Transformers 4.44.2
Software — Libraries	Supporting libraries	pandas, numpy, scikit-learn, Plotly, matplotlib, nltk, contractions, google-generativeai, openai (via OpenRouter), huggingface_hub (pinned <1.0)
Software — Development Environment	IDE / OS / tooling	Windows, Python 3.12.10, local virtual environment, Git + GitHub
Data	Source, size, format	GoEmotions + EmpatheticDialogues + ISEAR, merged and relabeled into 5 target classes (Bored, Confident, Confused, Curious, Frustrated); 770-row held-out validation set used for the decision-engine analysis.